Laboratory Work No. 2. Decision Trees and Random Forest.

After doing the Laboratory work No.2, fill the review table and in *.pdf form submit it to Moodle „Laboratory Work No. 2“.

Name, Surname: Denys Suntsov

Question No.		Leave the actual value or answer a question
General requirements		
	Provide information about files submitted to moodle:	
0.1	Report is submitted in pdf format	Yes
0.2	Scripts are submitted (*.py, *.ipynb, ...)	Yes
0.3	In laboratory report provided:	
0.3.1	Student‘s name, surname	Yes
0.3.2	Laboratory work number	Yes
0.3.3	Laboratory task	Yes
0.3.4	Theoretical background of the model	Yes
0.3.5	Strategy for training the model (description or scheme)	Yes
0.3.6	Training and testing results	Yes
0.3.7	Conclusions	Yes
Regression Decision Tree		
2.	Provide information about the implemented regression decision tree and its description in the report:	
2.1	What criterion was used for best feature selection in splitting?	SSE
2.2	What conditions for terminating tree construction were used?	*When max_depth was reached, min_samples_split was not satisfied, node was pure, min_leaf_size couldn’t be satisfied, no valid split improved splitting criterion*
2.3	The tree is designed to handle both categorical input features (without converting them to numerical values) and continuous ones;	Yes
2.4	for categorical features, separate branches are created for each category;	Yes
	for numerical features, five threshold values are tested;	Yes
	for binary features, two branches are created immediately.	Yes
2.5	The table shows 5 combinations of decision tree parameter values and the corresponding accuracy metrics of the implemented models for the training and validation datasets	Yes
2.6	Justification is provided for which set of parameters should be used.	Yes
2.7	The table shows 5 combinations of decision tree parameter values and the corresponding accuracy metrics of the models implemented by library functions for training and validation datasets	Yes
2.8	The results of the implemented models and those obtained using the library functions were compared	Yes
2.9	At least the first three layers of the best-fit tree are plotted or printed (criteria)	Yes

Classification Decision Tree		
3.	Provide information about the implemented classification decision tree and its description in the report:	
3.1	Continuous target variable is mapped to categorical by creating 5 classes	If Price < 571,000 → Very small If 571,000 ≤ Price < 721,000 → Small If 721,000 ≤ Price < 908,000 → Medium If 908,000 ≤ Price < 1,250,000 → Large If Price ≥ 1,250,000 → Very large
3.2	What criterion was used for best feature selection in splitting?	*Gini*
3.3	What conditions for terminating tree construction were used?	*Tree construction stops when maximum depth is reached, minimum sample requirements are not met, the node is pure, or no better split is possible.*
3.4	The tree is designed to handle both categorical input features (without converting them to numerical values) and continuous ones;	Yes
3.5	for categorical features, separate branches are created for each category;	Yes
	for numerical features, five threshold values are tested;	Yes
	for binary features, two branches are created immediately.	Yes
3.6	The table shows 5 combinations of decision tree parameter values and the corresponding accuracy metrics of the implemented models for the training and validation datasets	Yes
3.7	Justification is provided for which set of parameters should be used.	Yes
3.8	Confusion matrices of the selected model are provided for both training and validation datasets.	Yes
3.9	The table shows 5 combinations of decision tree parameter values and the corresponding accuracy metrics of the models implemented by library functions for training and validation datasets	Yes
3.10	The results of the implemented models and those obtained using the library functions were compared	Yes
3.11	At least the first three layers of the best-fit tree are plotted or printed (criteria)	Yes
Random Forest for Regression		
4.	Provide information about the implemented random forest and its description in the report:	
4.1	All data from the training dataset is used to construct one decision tree in the forest. (If No, write which part is used)	No. Each tree is trained on a bootstrap sample of the training dataset (random sampling with replacement).

4.2	All features of the training sample are used to construct one decision tree in the forest. (If No, enter how many features are used)	No, only √(total number of features), rounded down, are randomly considered at each split.
4.3	The report shows the results of cross-validation (rectangular plots, value tables, etc.) of k-subsets (k value from 5 to 10) when at least 27 different parameter combinations are used to create a forest (3 tree numbers, 3 values of two tree parameters each)	Yes n_estimators_list = [3, 5, 10] max_depth_list = [3, 5, 8] min_samples_split_list = [5, 10, 15]
4.4	The report provides a comment on the results of the cross-validation performed in 4.3, a comment on the best data set, and a comment on how the model performance depends on the number of trees.	Yes
4.5	The report contains the results of cross-validation if standard library scikit-learn is used to generate random forest (RandomForestRegressor).	Yes
4.6	A random forest model was created with the parameters selected during the study and trained with the training dataset.	Yes
4.7	The generated random forest model is applied to the records of the original format.	Yes
4.8	The report provides model accuracy metrics for the test dataset.	Yes
4.9	Display at least 10 test sample records with predicted and actual values in the report.	Yes
4.10	There is an option to enter the values of a single record in the original format and get the value predicted by the model without training.	Yes